WEEK 08 • IS321 ENTERPRISE ARCHITECTURE

Business Continuity Planning

Keeping the Enterprise Running When Everything Goes Wrong

TODAY'S ROADMAP

What We'll Cover

01	**What Is BCP?** – definition	A proactive process for identifying and protecting an enterprise's critical functions
02	**Why BCP, and Who Governs It** – stakeholders involved	The business case, plus senior management's specific responsibilities
03	**The BCP Life Cycle & Phases** (complementary models)	Two complementary models for how a continuity plan actually gets built
04	**Maintaining the Plan** – refining the plan over time as requirements evolve	BCP is a process, not a one-time project
05	**Disaster Recovery**	Cost curves, hot sites, and cold sites
06	**AI for Business Continuity Planning**	Incident prediction, DR optimization, and automated monitoring

↓ integrating AI

↓ reducing costs incurred

What Is Business Continuity Planning?

> *A proactive process that identifies and prioritizes an enterprise's critical functions and the likely threats to them — sustained through regular personnel training, plan testing, and maintenance.*
>
> — Definition of Business Continuity Planning

Set by senior management

Worst-Case Scenarios

Senior management defines the worst-case scenarios that anchor the whole planning process.

Funding & Staffing

Senior management approves the funding and staffing of the BCP program itself.

Operating Under Limits

Every BCP discipline and technique operates within the circumstances and limits management sets.

MAKING THE CASE

Why Should We Do BCP?

→ to avoid disruption and ensure organization operates smoothly despite major crisis

It's the Right Thing to Do

For the business itself, its staff, and the communities it operates in.

Regulatory Compliance

Ensures compliance with ever-increasing regulatory requirements across industries.

It enhances our ability to avoid:

1. Interruptions to business affairs
2. Financial losses
3. bearing Regulatory fines
4. Damage to equipment

WHO'S RESPONSIBLE

BCP Governance — Senior Management's Role

The organization's senior management team is responsible for overseeing the entire BCP process:

1. Establish policy for managing and controlling identified risks
2. Allocate knowledgeable personnel and sufficient financial resources
3. Ensure the BCP is reviewed and approved at least annually
4. Ensure employees are trained and aware of their BCP roles
5. Ensure the BCP is regularly tested
6. Review BCP testing programs and results on a regular basis
7. Ensure the BCP is continually updated to reflect the current environment

APPROACH

How BCP Actually Works

A Process, Not a Project

Business continuity is treated as an ongoing process, never a one-time deliverable.

Proven, Certifiable Discipline

Methods are based on established, certifiable BCP disciplines — not improvised from scratch.

Starts Simple, Grows Sophisticated

Plans begin simply and increase in sophistication as teams gain confidence and experience.

Governed by Senior Management

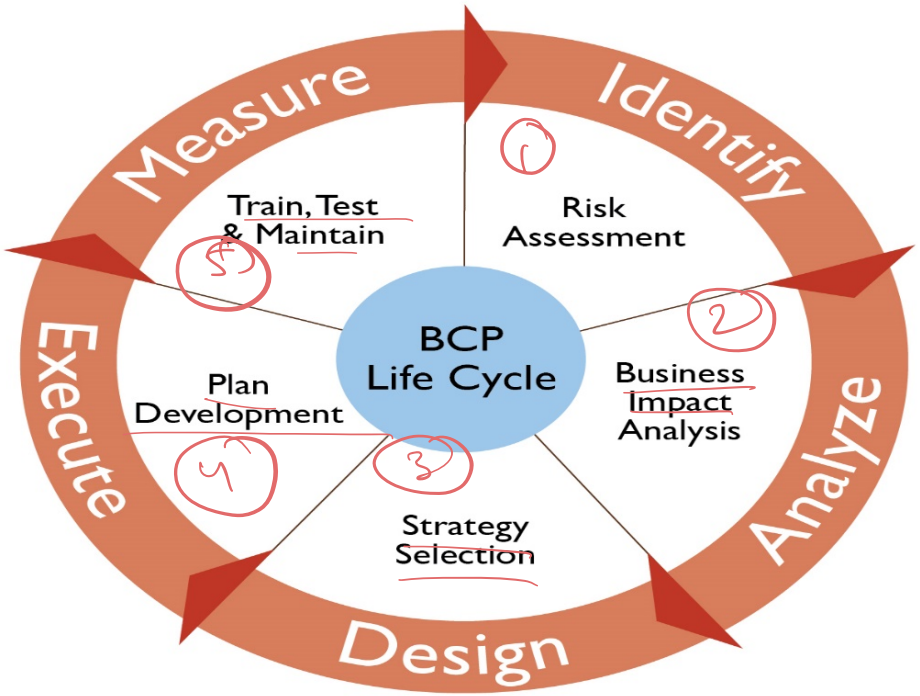
Governance is provided for the whole system by senior management, as covered on the previous slide.

Interwoven with Emergency Management

Established emergency management activities and protocols are woven directly into the BCP process.

MODEL 1 OF 2

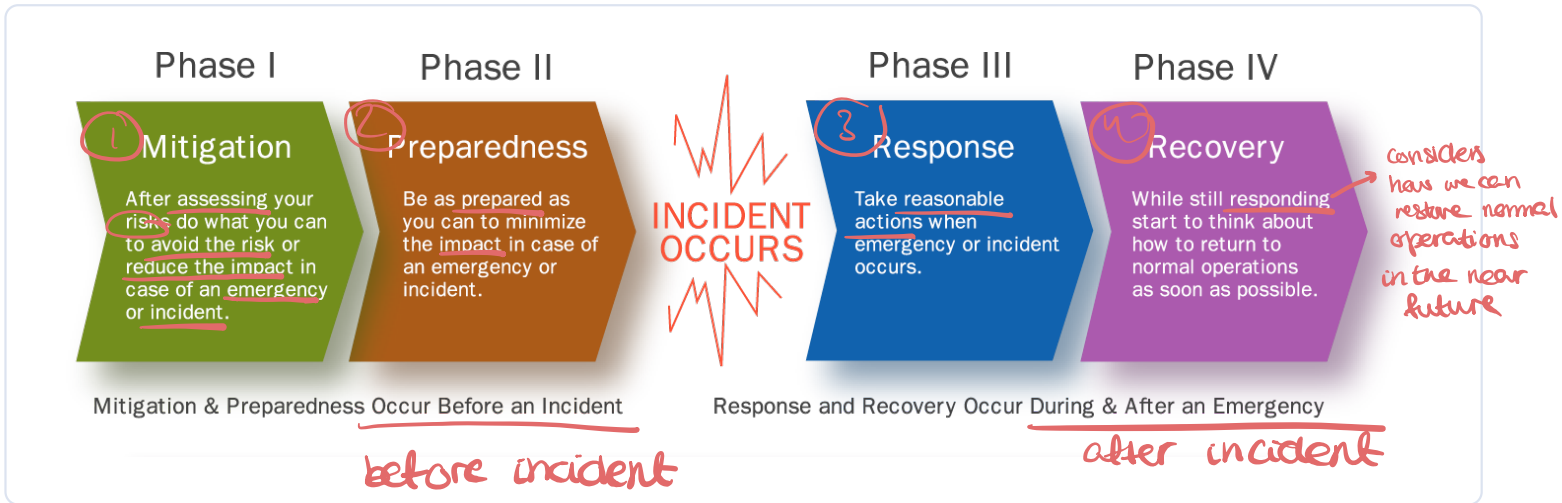
The BCP Life Cycle



A continuous, five-stage cycle: Identify (risk assessment) → Analyze (business impact) → Design (strategy selection, plan development) → Execute → Measure (train, test, maintain) → back to Identify.

MODEL 2 OF 2

Phases of Continuity Planning



Mitigation and Preparedness happen before an incident; Response and Recovery happen during and after. Note this is a timeline, not a cycle — a different lens on the same underlying work as the Life Cycle model.

ONGOING WORK

How Will We Maintain It?

BCP requires participation from all levels of the organization — maintenance is everyone's job, not just senior management's.

- Create or update business continuity plans
- Maintain current, easily accessible emergency contact information
- Test business continuity plans
- Assign key players to emergency response teams
- Establish alternate work arrangements
- Know the organization's emergency management procedures
- Save all work products to the department's shared server
- Store a disaster kit (“grab and go”) with items needed during an emergency

Disaster Recovery Best Practices

Mind the Enterprise Architecture

Disaster recovery planning should be informed by the EA repository — not treated as a separate, disconnected exercise.

Monitor Network Quality

Track the quality of computer networks providing data on power supply and demand.

Ensure Fast Restoration

Make sure networks can be restored quickly in the event of downtime.

Set Up Recovery Plans

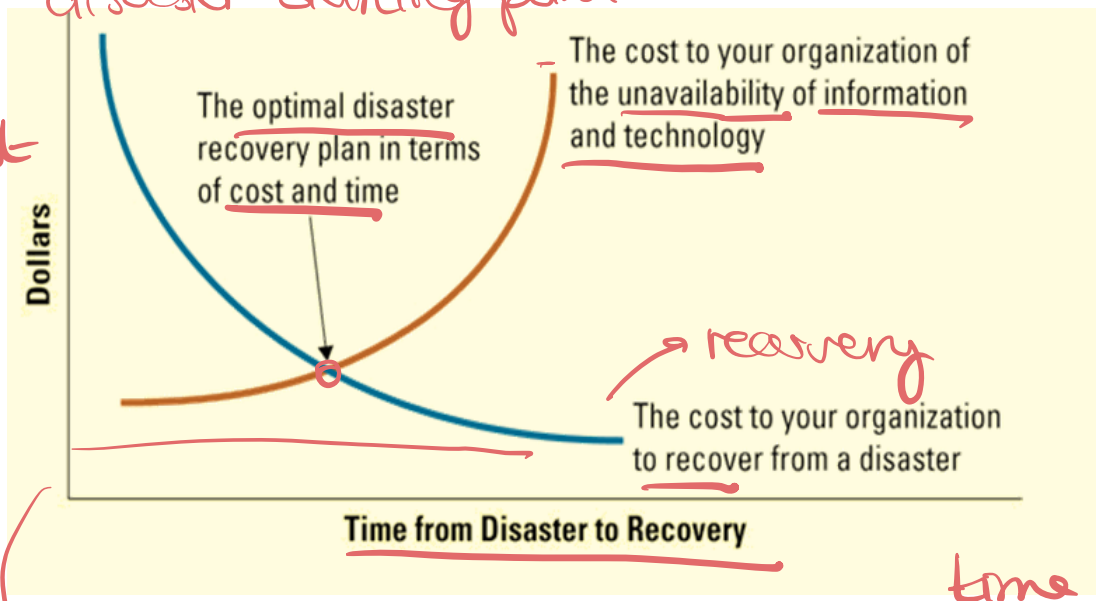
Formal disaster recovery plans should exist before they're needed, not written during a crisis.

Train Staff Adequately

Adequate staff training ensures the plan can actually be executed under pressure.

The Disaster Recovery Cost Curve

Disaster recovery plan: a detailed process for recovering information or an IT system after a catastrophic event (fire, flood, etc.). The cost curve below is how organizations find the optimal investment level.



Hot Sites vs. Cold Sites

Hot Site

A separate, fully equipped facility the company can move to immediately after a disaster and resume business right away.

Cold Site

A separate facility with no computer equipment — just a place employees can relocate to after a disaster.

NEW SECTION

AI for Business Continuity Planning

Everything above describes how BCP has worked for decades. AI is now changing how enterprises anticipate incidents, optimize recovery, and monitor continuity in real time.



Incident Prediction



Disaster-Recovery Optimization



Automated Continuity Monitoring

AI for Incident Prediction



The BCP Life Cycle's "Identify" stage traditionally relies on periodic risk assessments. AI-based anomaly detection can continuously monitor operational data — network traffic, equipment sensors, weather feeds — to flag early warning signs of a disruption before it becomes an incident.

EXAMPLE

A model trained on server performance data flags an unusual pattern of intermittent failures three days before a full outage would have occurred — giving the BCP team time to shift to preparedness mode early.

AI for Disaster-Recovery Optimization

Recall the disaster recovery cost curve: organizations look for the crossing point between unavailability cost and recovery cost. AI-assisted scenario modeling can recompute that optimal point dynamically — as systems, data volumes, and dependencies change — rather than relying on a static, once-a-year estimate.

EXAMPLE

As a university's online-degree platform (Week 3's running example) grows, an AI model recalculates that the cost curve's crossing point has shifted — recommending an upgrade from a cold site to a warm site for that specific system.

AI for Automated Continuity Monitoring

The "Measure" stage of the BCP Life Cycle — train, test, and maintain — has traditionally meant scheduled, periodic testing. AI-driven monitoring tools can continuously verify that continuity controls (backups, failover systems, alternate sites) are actually functional, rather than waiting for the next scheduled test to find out.

EXAMPLE

An automated monitoring system detects that a backup job silently failed for the past two nights and alerts the BCP team immediately — instead of the gap being discovered during next quarter's scheduled test.

CHAPTER SUMMARY

Key Takeaways

- BCP is a proactive process — identifying and prioritizing critical functions and threats — governed and funded by senior management.
- The BCP Life Cycle (5 continuous stages) and the Four-Phase model (Mitigation/Preparedness/Response/Recovery) describe the same discipline from two complementary angles.
- BCP maintenance is everyone's responsibility, not a one-time senior-management task.
- Disaster recovery has its own economics — the cost curve balances unavailability cost against recovery cost, and hot/cold sites sit at opposite ends of that trade-off.
- AI adds three capabilities on top of traditional BCP: incident prediction, disaster-recovery optimization, and automated continuity monitoring.

COMING UP NEXT

Further Reading & Next Session

Reading Material

- Bernard, S. (2012). An Introduction to Enterprise Architecture (3rd ed.), Chapter 8 — Primary text
- PSU IS321 Syllabus — AI and EA Tools policy
- Week 3's five-level model (referenced in the DR-optimization example above)

Next: Week 9

Linking EA and Strategy, and Linking EA and Business Planning — how strategic goals become documented business activities. Mapped to CLO5.

Questions?